

Notes on Quantitative Analysis (Topics)

Hypothesis Test for a Single Proportion:

When you want to test a hypothesis about a population proportion (like the proportion of people who prefer brand A over brand B).

Possible Statistical Test:

Z-test for a Single Proportion: This test is commonly used when you have a large sample size (typically $n > 30$) and you want to test a hypothesis about a single population proportion. It assesses whether the observed proportion differs significantly from a hypothesized population proportion.

Hypothesis Test for the Difference Between Two Proportions:

When you want to compare two population proportions (e.g., comparing the proportion of men who prefer brand A to the proportion of women who prefer brand A).

Possible Statistical Test:

Z-test for the Difference Between Two Proportions: This test is appropriate when you have two independent samples and want to compare the proportions from each sample. It helps determine if there is a significant difference between the proportions. The test statistic follows a standard normal distribution under the null hypothesis.

Variance:

Variance is a measure of the spread or dispersion of a set of data points. It's calculated as the average of the squared differences from the mean. In other words, it gives you an idea of how much individual data points deviate from the mean of the dataset.

Equality of Variance:

Equality of variance refers to the assumption that the variance is the same across different groups or conditions in a statistical test. In simpler terms, it means that the spread of data points is similar across all groups being compared. This assumption is particularly important in tests such as the analysis of variance (ANOVA) and the t-test. Violations of this assumption can lead to inflated Type I error rates (false positives) or decreased power in detecting true differences.

Correlation Coefficient:

The correlation coefficient is a statistical measure of the strength of a linear relationship between two variables. Its values can range from “-1 to 1”. A correlation coefficient of -1 describes a perfect negative, or inverse, correlation, with values in one series rising as those in the other decline, and vice versa. A coefficient of 1 shows a perfect positive correlation, or a direct relationship. A correlation coefficient of 0 means there is no linear relationship.

Limitations of Hypothesis Testing:

Hypothesis testing has some limitations that researchers should be aware of:

- *It cannot prove or establish the truth:* Hypothesis testing provides evidence to support or reject a hypothesis, but it cannot confirm the absolute truth of the research question.
- *Results are sample-specific:* Hypothesis testing is based on analyzing a sample from a population, and the conclusions drawn are specific to that particular sample.
- *Possible errors:* During hypothesis testing, there is a chance of committing type I error (rejecting a true null hypothesis) or type II error (failing to reject a false null hypothesis).
- *Assumptions and requirements:* Different tests have specific assumptions and requirements that must be met to accurately interpret results.

